

ELEC 5660: Introduction to Aerial Robotics

Project 3: Phase 3 (Group Work)

Released: Apr. 28, 2026

Due: May 22, 2026

1. Project Work

In this lab assignment, you need to integrate the whole system onboard. Using the information from the IMU and camera, your visual estimator and the EKF compute the state of the quadrotor. The drone will use this state for feedback control and execute trajectories computed by your path planner and your trajectory generator.

In the real flight, we do not provide the tag due to the system environment. You need to check your code on the ground and ensure it can run without the tag.

If you finish the Extended Kalman Filter well, there should not be theoretical difficulties in this project. The remaining work is all about system integration and testing. Please be careful with the world frame, especially how state estimation is handled in your visualization. Your EKF calculates the pose of the quadrotor IMU in the world frame. The control node requires the pose of the quadrotor IMU in the Opti-Track world frame. You should compare your pose and velocity with the Motion Capture system before flight.

Be careful with your drone. Before autonomous flight, you should first do some tests with the drone in hand to see whether the filter and state estimator are good. Almost all bugs can be found offline, so please think carefully about what the problem is and how to deal with it. Do not just make it crash repeatedly. If you cannot find the bug and try 100 times, you may crash 100 times.

2. Submission

When you finish the assignment, you may submit your code, documents, and video on Canvas before **May 22, 2026, 23:59:00**.

The project name for this assignment should be titled:

`GroupX-Proj3Phase3`

where `X` is the number of your group.

Your submission should be a `.zip` file containing:

1. A maximum 4-page document including figures plotted by `rqt_plot` or `rviz`, descriptions of your implementation, and any other things we should be aware of;
2. A demonstration video;
3. Your code and all files needed to run your code.